Journal of Electrical Systems and Information Technology

RESEARCH Open Access

SheepFormers: vision transformer-based sheep breed classification on small datasets for smart farming

Galib Muhammad Shahriar Himel[1], Md. Masudul Islam[2*] and Mijanur Rahaman[2]

*Correspondence:
Md. Masudul Islam
masudulislam11@gmail.com
[1]American International University-Bangladesh, Dhaka, Bangladesh
[2]Bangladesh University of Business and Technology, Dhaka, Bangladesh

Abstract

Accurate sheep breed classification is critical for modern livestock management, supporting decisions on breeding, productivity, and profitability in the farming industry. Traditional methods based on visual inspection or body measurements are often subjective, time-consuming, and unsuitable for large-scale operations. To address these challenges, this study proposes SheepFormers, a Vision Transformer (ViT)-based framework for automatic sheep breed identification, specifically designed to operate effectively even with small datasets. A balanced dataset of 1,680 sheep face images representing four breeds was preprocessed and used to evaluate multiple transformer variants, including ViT-Google, ViT-MAE, ViT-VAN, BEiT, ViT-ResNet50, and DiT. Through systematic hyperparameter optimization and ablation studies covering epochs, batch size, learning rate, positional encoding, patch size, data augmentation, and 5-fold cross-validation, the ViT-Google model achieved the best performance with 98.21% validation accuracy and an Area Under the Curve (AUC) of 0.9987, outperforming both traditional CNN architectures and ensemble baselines reported in previous studies. Despite these strong results, the study is limited by the relatively small dataset size and controlled image acquisition conditions, which may affect generalization under diverse real-world farm environments. To ensure practical applicability, the trained model was deployed into a user-friendly mobile application, enabling farmers, buyers, and sellers to classify breeds in real time. These findings demonstrate that Vision Transformers can achieve state-of-the-art performance even with limited agricultural datasets, marking a significant advancement in smart farming technologies and precision livestock management.

Keywords Smart agriculture, Sheep breed classification, Image processing, Vision transformer, Image classification

Introduction

Artificial intelligence (AI) and computer vision have demonstrated broad applicability across diverse fields such as biomedical imaging [1–3] and agriculture, supporting critical tasks of identification, classification, and forecasting. In the livestock sector, AI has

© The Author(s) 2026. **Open Access** This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

been applied for breed identification in sheep, cattle, and birds, along with applications in facial expression analysis, vocal pattern recognition, foraging behaviour studies, plant identification [4], and sex differentiation in animal species [5].

Accurate classification of sheep breeds is crucial to enhancing productivity, disease management, and informed breeding decisions. The American Sheep Industry Association recognizes 47 distinct sheep breeds, classified by traits such as wool length, hair pattern, meat quality, and wool quality [6]. For instance, Marino sheep, originally from Spain, are valued for their fine fleece and adaptability to dry climates, though they yield fewer lambs [6]. Dorset sheep, from southern England, are medium-sized, resilient, and well-suited for varied environments, producing high-quality meat [6]. Suffolk sheep are known for their large size, rapid growth, and superior meat quality, but they face reproductive challenges compared to Dorset [6]. In practice, farmers must optimize profitability by improving output from the less productive portion of flocks [7]. Yet, conventional assessments often emphasize live weight alone, overlooking key factors such as wool yield and gut fullness [8]. Furthermore, managing multiple breeds on farms complicates decisions, since meat and milk yields vary by breed [6,36].

Previous research has explored machine learning approaches for sheep breed classification [9–12], but subtle inter-breed similarities remain a challenge. To address this, the present study leverages Vision Transformers (ViT), alongside models such as ResNet50, Visual Attention Network (VAN) [14], Masked Autoencoder (MAE) [15], Document Image Transformer (DiT) [16], and BEiT [17]. Specifically, Google's ViT (patch 32) [13] was employed to enhance classification performance, reducing dependency on manual expertise or costly DNA testing [18].

Although ViTs are typically designed to excel with large-scale datasets due to their high parameter count and reliance on extensive training samples, our study demonstrates their effectiveness in a small-data scenario. We applied ViT to a relatively limited sheep breed dataset consisting of 1,680 images, showing that with careful preprocessing, hyperparameter optimization, and supervised fine-tuning, ViTs can achieve state-of-the-art accuracy even without massive data. This highlights the adaptability of transformer architectures and their potential for agricultural applications where large annotated datasets are often scarce.

This research contributes to advancing smart agriculture by demonstrating the viability of ViT in automatic sheep breed identification, enabling greater precision and efficiency in farming [19]. To ensure real-world applicability, the proposed model was integrated into the *SheepFormers* mobile app, which uses farm-installed cameras to capture images and automatically identify breeds. Results are delivered directly to farmers, buyers, and sellers through the app, supporting accessible breed information and farm management. As illustrated in Fig. 1, the system not only classifies sheep breeds but also extends to smart farm management by offering environment control, breed-specific food recommendations, and medication suggestions, thereby enhancing productivity and decision-making efficiency [34]. This study makes the following contributions:

i. Sheep Breed Classification on a Small Sized Dataset.
ii. Demonstrating robust performance despite limited data by employing ViT and systematic optimization.
iii. Introducing Vision Transformers (ViT) for State-of-the-Art Accuracy and achieving superior results compared to conventional deep learning baselines.

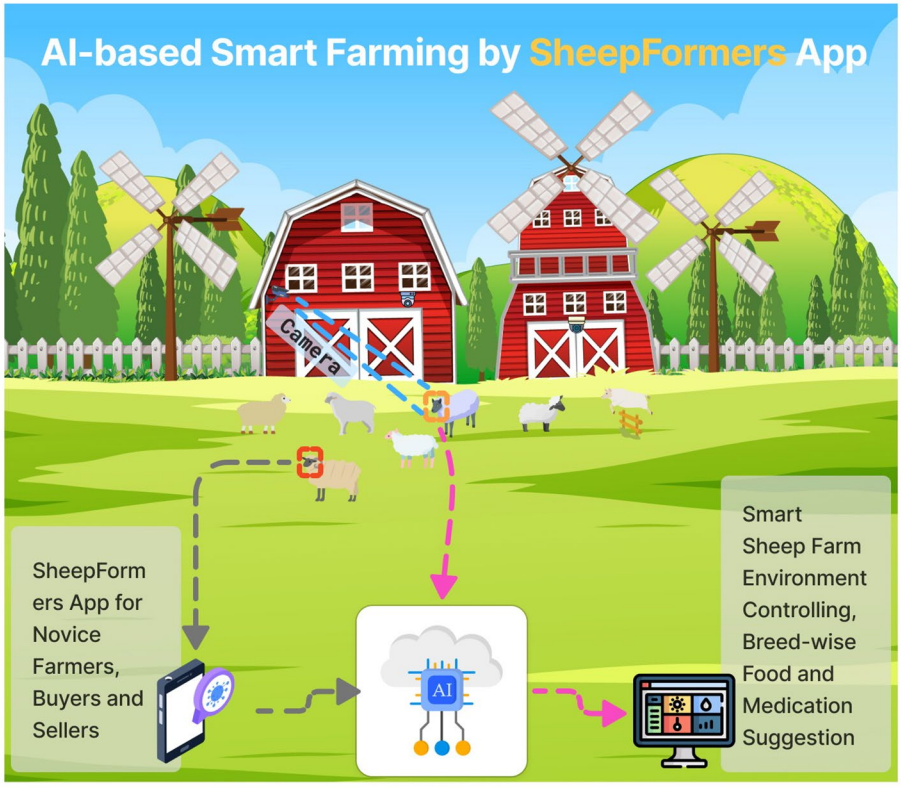


Fig. 1 Overview of the SheepFormers mobile app for smart farming. This conceptual illustration was created by the authors to demonstrate the system workflow and is inspired by real-world farm monitoring setups described in prior smart agriculture studies

iv. Developing a Smart Mobile Application and deploying the trained model into a real-time, user-friendly mobile app for farmers to enable accessible and practical sheep breed identification.

Related works

AI and CV techniques have been widely applied for prediction, categorization, and identification tasks, including agricultural animal management [8]. In sheep farming, prior CV-based research has focused on parameters like body condition, lameness, and cleanliness [18–21]. However, accurate breed identification remains challenging due to issues such as uniformly colored flocks, deformable body shapes, and the masking effect of wool [22–26].

Earlier methods relied on body measurements, achieving moderate accuracy (e.g., Carneiro et al. with 86.2% [27]). Later, hybrid machine learning approaches improved performance, with Salama et al. reporting 98% accuracy using deep learning with Bayesian optimization [28]. Transfer learning has also shown promise: Agrawal et al. achieved strong results with ResNet50V2 ensembles [29], while Hitelman et al. developed a biometric system with 97% accuracy [31]. Himel et al. advanced this further with ensemble learning, reaching 97.92% [32]. Yuan et al. proposed SheepFaceRec combining MTCNN and FaceNet, but noted limitations in dataset diversity and comparative analysis [33]. Overall, existing research demonstrates encouraging results but also reveals persistent gaps:

i. Small and imbalanced datasets often restrict generalization.
ii. High false positive rates occur for certain breeds despite high reported accuracy.
iii. Over-reliance on CNNs and transfer learning without exploring transformer-based models.
iv. Limited real-world usability, as most studies stop at accuracy metrics without deployment into farmer-friendly tools.

Our work addresses these gaps in three significant ways. First, we demonstrate reliable sheep breed classification on a small dataset of 1,680 images by leveraging ViT with careful preprocessing, optimization, and external validation. Second, we introduce ViT into this domain and show that it achieves state-of-the-art accuracy of 98.21%, outperforming conventional CNNs and reducing false positives. Finally, we bridge the gap between research and practice by developing SheepFormers, a smart mobile application, which deploys the trained model for real-time breed identification and integrates farm management functionalities such as environment control, nutrition guidance, and medication recommendations.

Methodology and implementation

This study employed a structured pipeline to develop an automated sheep breed classification system using ViT. The workflow included dataset preparation, preprocessing, and supervised training of ViT models with systematic hyperparameter tuning. Model performance was validated across multiple ViT variants and further tested on real-life data. Finally, the optimized model was deployed in a mobile application to provide farmers with an accessible, real-time tool for breed identification.

Dataset description

We made use of the 1680 image Sheep Breed dataset, where 420 photos were assigned to each breed. Each picture is 156×181 pixels in size. In the training phase, 224×224 was the new image size. Images were resized from 156×181 to 224×224 pixels to match the fixed input size required by pre-trained Vision Transformer architectures, enabling effective patch embedding and transfer learning while maintaining consistency across all models. The pictures were originally collected from the South-West part of Western Australia, by S. Abu Jwade [18]. As shown in Table 1, the data set was then split into three segments: 80% for the training photos (1344), 10% for the validation images (168), and an additional 10% for the test images (168).

Vision transformer for classification

This study aims to enhance sheep breed classification on farms by employing a fine-tuned ViT trained on a relatively small image dataset. Raw sheep images were preprocessed by resizing them to 224×224 pixels; data augmentation was not required due to

Table 1 Dataset description

Total images	Train images	Val images	Test images
Marino	336	42	42
Poll Dorset	336	42	42
Suffolk	336	42	42
White Suffolk	336	42	42
Grand Total	**1344**	**168**	**168**

sufficient dataset size. The ViT model takes these images as input, splits them into fixed-size patches, and converts each patch into lower-dimensional embeddings, which serve as Transformer tokens. Positional embeddings provide spatial information, while the transformer encoder—comprising layers of self-attention and feedforward networks—captures both local and global dependencies. Multi-head attention enables the model to focus on multiple image regions simultaneously, generating robust feature representations. Layer normalization enhances stability and generalization, and a classification head with SoftMax activation outputs the final breed probabilities.

We employ Google's ViT model for supervised training, achieving high accuracy in differentiating between sheep breeds. Figure 2 illustrates the overall architecture of the proposed system. Algorithm 1 is based on the standard Vision Transformer formulation proposed by Dosovitskiy et al. [13], with adaptations for supervised sheep breed classification.

Hyperparameter optimization

To systematically tune the ViT model, we experimented with different epochs, batch sizes, learning rates, optimizers, and loss functions. Training was run for 50, 75, 100, and 125 epochs; while good results appeared at 50 epochs, stability was only achieved around 95–100 epochs, with no significant improvements beyond that point. For batch size, values of 8, 16, 32, and 64 were tested, and although accuracy remained similar across settings, a batch size of 32 provided the most efficient GPU utilization. Regarding learning rate, we tested values around the standard ViT range (1e-4 to 5e-5). A rate of 0.0001 destabilized convergence, while lowering it to 0.00002 stabilized training and yielded the best results, albeit with slower convergence. Further reduction offered no additional benefit. Finally, using constant epochs, batch size, and learning rate, we compared optimizers (Adam, Adamax, RMSProp) and loss functions (categorical cross-entropy, Kullback–Leibler divergence). The best performance was achieved with the Adam optimizer combined with categorical cross-entropy loss, as summarized in Table 2.

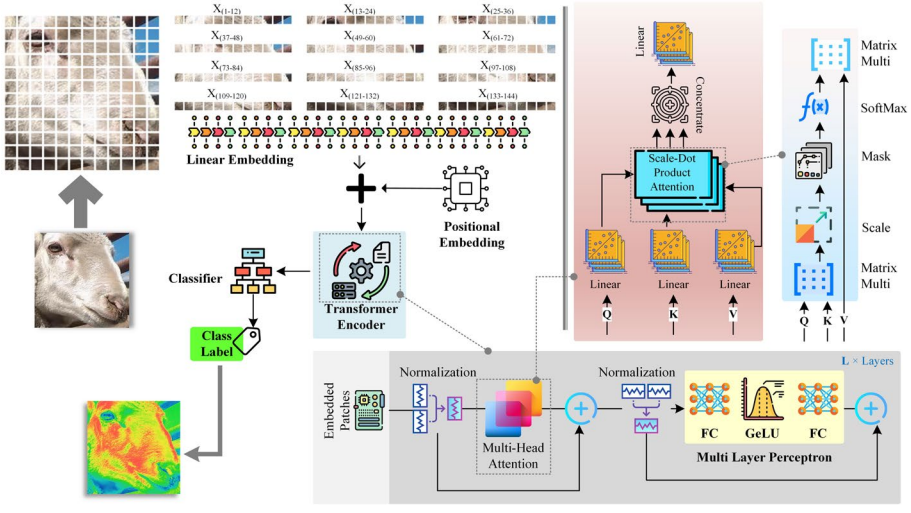


Fig. 2 General architecture of the proposed Vision Transformer model [13]. The pipeline includes image preprocessing, patch embedding, multi-head self-attention, and a classification head. This structure allows the model to capture fine-grained features critical for distinguishing similar sheep breeds

Table 2 Hyperparameter optimization for classification using ViT

Hyperparameter	Optimization Space
Epoch	100
Batch size	32
Learning rate	0.00002
Optimizer	Adam
Loss function	Categorical Cross Entropy Loss

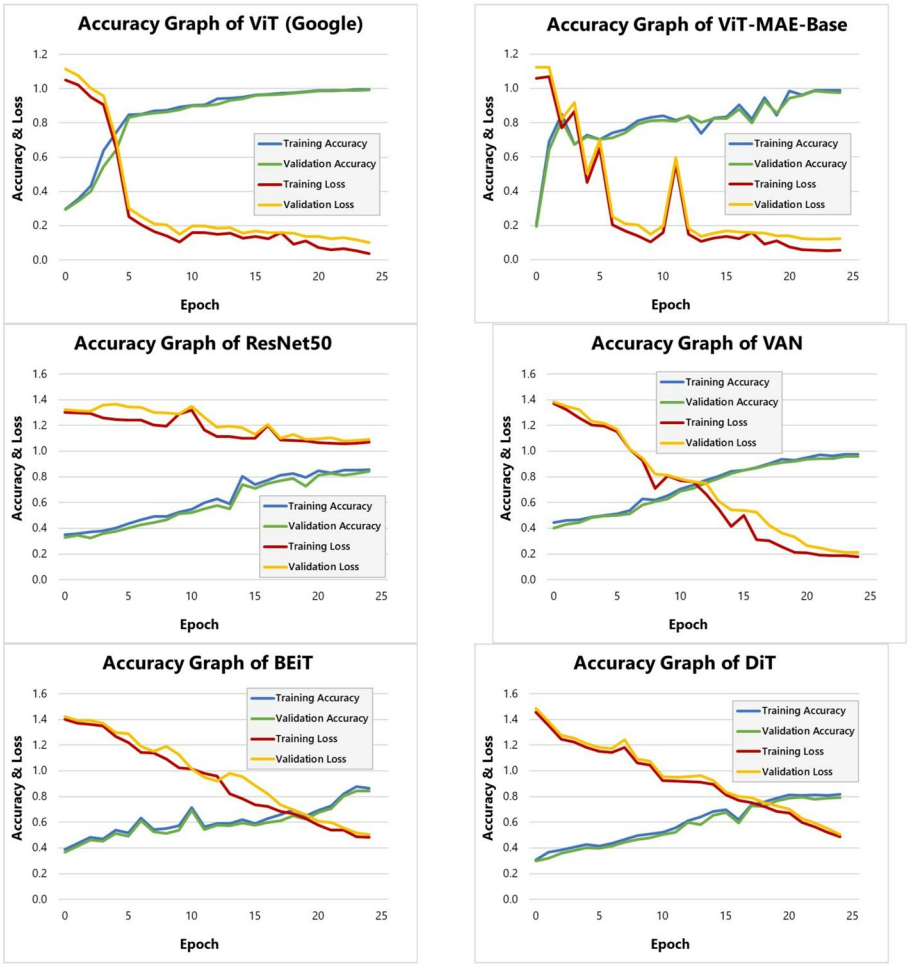


Fig. 3 Training and validation accuracy/loss curves for different ViT variants. The curves show that ViT-Google converges more stably and achieves the highest validation accuracy (98.21%), outperforming other models

Results and discussion

In this section, we present the results of our study, including graphical illustrations and comparative analyses. To evaluate the generalizability of our model, it was further tested on an external unseen dataset. Finally, we developed a real-time Android application capable of recognizing sheep breeds.

Accuracy

Figure 3 illustrates the loss and accuracy curves of the various ViT models. It is evident from the graph that the training patterns are mostly similar across all models, albeit with some slight variations. It provides a comparison among different types of ViT models.

Table 3 Comparison of various version of ViT models

Model	Training Loss	Training Accuracy	Validation Loss	Validation Accuracy	Test Accuracy
ViT-Google*	0.03677	0.99702	0.10134	0.99405	0.98214
ViT-MAE	0.05766	0.98810	0.12223	0.97619	0.97619
ViT-ResNet50	1.07253	0.85714	1.09253	0.84226	0.83929
ViT-VAN	0.17871	0.97619	0.21113	0.96131	0.95238
ViT-BEiT	0.48391	0.86607	0.50391	0.48391	0.84524
ViT-DiT	0.48579	0.81845	0.50579	0.79167	0.79167

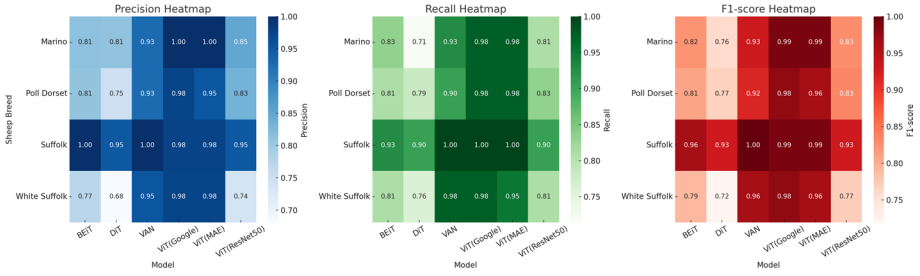


Fig. 4 Classification report heatmaps for F1-scores across breeds and models. Darker red shading indicates stronger performance; ViT-Google and ViT-MAE achieved consistently high scores across all classes, while DiT underperformed

The accuracy obtained for the following ViT-Google, ViT-MAE, ViT-ResNet50, ViT-VAN, ViT-BEiT, and ViT-DiT models is mentioned in detail in Table 3.

Classification report

For evaluating classification performance, we used the F1-score, which balances precision and recall into a single metric. Precision measures the correctness of positive predictions, while recall indicates how well true positives are captured. The F1-score is particularly important in our study as it provides a fair assessment of model efficiency, especially when dealing with potential class imbalance. Figure 4 illustrates the classification report heatmaps of all applied ViT models.

Each sheep breed exhibits subtle similarities, except for 'Suffolk,' which stands out due to its distinctive facial coloration. ViT-Google and VAN, as well as ViT-MAE and BEiT, achieve perfect identification accuracy for Marino and Suffolk breeds, respectively. However, the performance of other models in identifying the Marino breed is notably poor. Conversely, the Suffolk breed demonstrates consistently decent precision rates across all models. 'Poll Dorset' breed exhibits significantly low precision rates across all models except for Google's ViT model. ViT-ResNet50 and DiT models show notably poor performance in individual breed identification. The primary facial features extracted from images of each sheep breed using Google's ViT model approach are depicted in Fig. 5 via GradCAM and Guided-GradCAM.

Confusion matrix

Figure 6 shows the confusion matrices for the models evaluated. Upon examining these matrices, it is evident that in instances of misclassification, the false negative rate surpasses the false positive cases for all models. Notably, the overall test accuracy of ViT Google is higher than that of the other models. From Fig. 7, we can see that the ROC score or AUC of ViT-Google is 99.87%. ViT(Google) and ViT(MAE) demonstrate almost

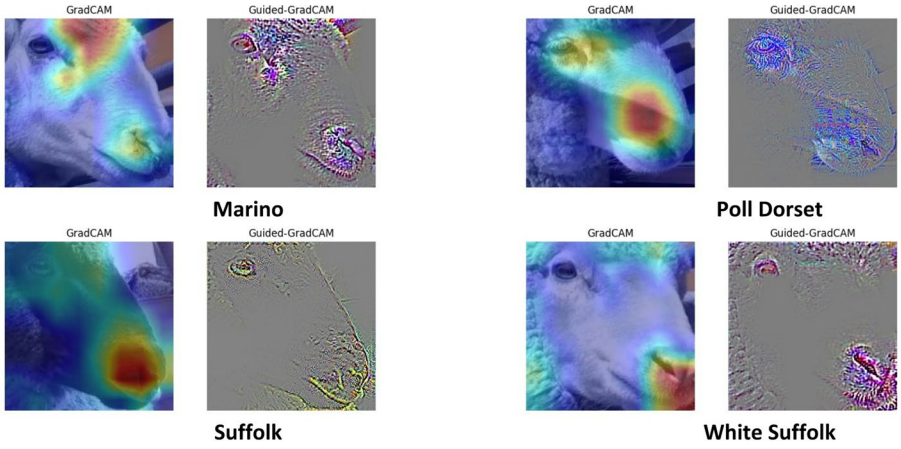


Fig. 5 Example of GradCAM and Guided-GradCAM Interpretation of Sheep Breed Images

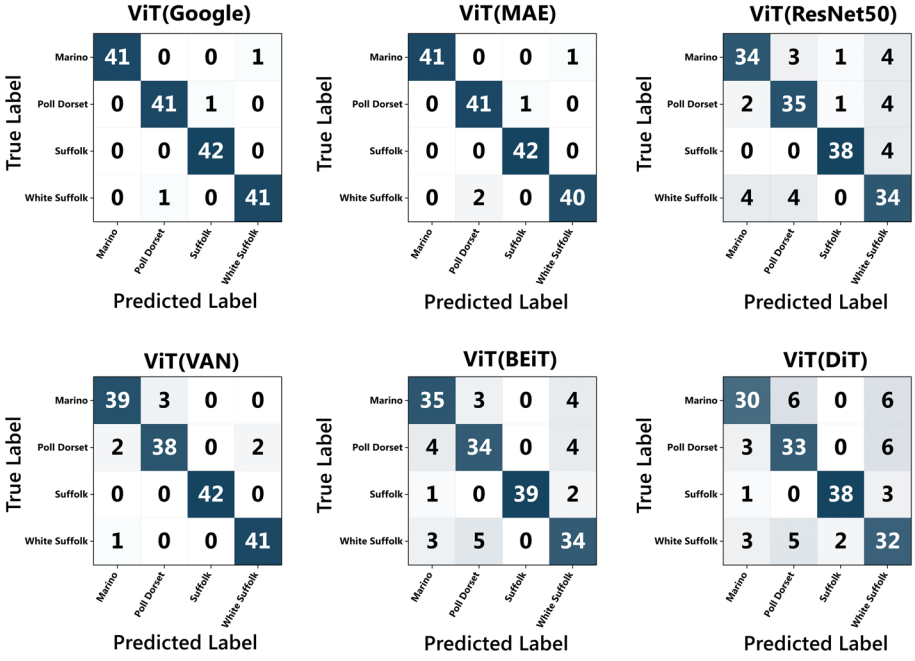


Fig. 6 Confusion matrix of the best-performing ViT-Google model. The results confirm high accuracy across all breeds, with minimal misclassifications, especially in visually similar Suffolk and White Suffolk classes

perfect classification capabilities, with minimal misclassifications across the breeds; for instance, both models misclassify only one White Suffolk sample as Marino. In contrast, ViT(ResNet50) and ViT(DiT) exhibit slightly poorer performance, with more significant errors, particularly for the Marino and White Suffolk breeds. Notably, ViT(DiT) struggles the most, misclassifying up to six Marino samples and five White Suffolk samples, indicating difficulty in distinguishing these breeds. While ViT(VAN) and ViT(BEiT) perform reasonably well, they still show some confusion, especially between Poll Dorset and Marino. Overall, ViT(Google) and ViT(MAE) are the top performers with only one or two misclassifications each, while ViT(DiT) faces the highest error rate across the models, particularly with White Suffolk and Marino. From this experiment, we can observe

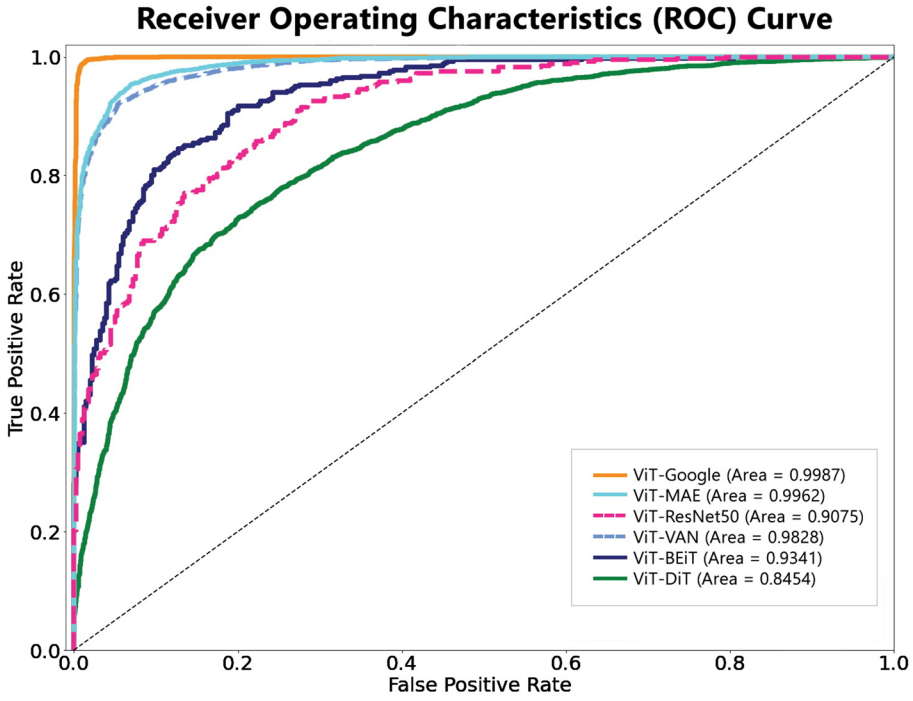


Fig. 7 ROC curves for all ViT models. The ViT-Google model achieved the highest AUC (0.9987), demonstrating superior discriminative ability compared to other transformer variants

Table 4 Baseline comparison summary

Model	Best Test Accuracy	Best ROC	Optimizer + Loss
Xception	84.7%	0.9684	RMSProp + CCE
VGG16	74.5%	0.9363	Adam + KLD
InceptionV3	79.9%	0.9451	Adam + KLD
InceptionResNetV2	81.5%	0.9368	Adam + KLD
DenseNet121	80.6%	0.9528	RMSProp + CCE

that both the false positive and false negative rates are reduced in comparison with the previous models.

Baseline model comparison

To establish strong baselines, we evaluated several popular CNN architectures (Xception, VGG16, InceptionV3, InceptionResNetV2, DenseNet121) under different parameter combinations of optimizers (Adam, Adamax, RMSProp) and loss functions (categorical cross-entropy, Kullback–Leibler divergence). A constant learning rate of 0.0001 was maintained for fair comparison. A full set of baseline results under different optimizers and loss functions is provided in **Appendix A**, Table A1 which is adapted from our previous work [32]. For clarity, Table 4 summarizes only the best-performing configuration of each model.

External dataset validation

In addition to the original dataset, we have gathered 80 images (20 images in each category) from different offline and online sources to examine the models' suitability and capability of generalizing for practical use. We have discovered that our models perform



Fig. 8 Example Images of Self-collected Dataset

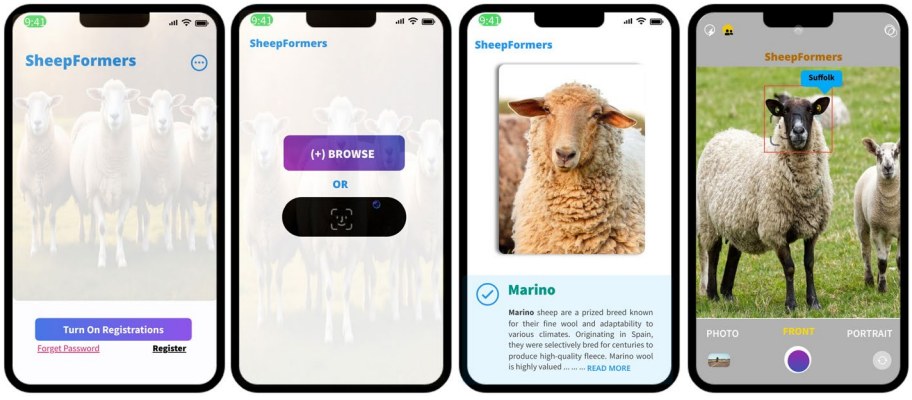


Fig. 9 Interfaces of SheepFormers Smart App

noticeably well on an entirely new test dataset. A sample of photos from a self-collected dataset is shown in Fig. 8.

Using our custom dataset, we have tested Google's ViT model trained on our initial dataset we got 97.5% accuracy which demonstrates the capability of the model in terms of other real-life data.

Model deployment and app interfaces

To make machine learning models usable by individuals lacking technical skills, it's crucial to convert the TensorFlow model into TensorFlow Lite, as smartphones cannot directly run TensorFlow models. Our application's development took place on the Android Studio platform, utilizing the Dart programming language and the Flutter framework. This mobile app lets users choose images from storage or capture them using the camera. Once the desired image is acquired, the app promptly provides the classification result. Figure 9 illustrates some sample images of the user interface.

Comparison study

The outcomes of our proposed approach were benchmarked against prior studies conducted on the same dataset, as summarized in Table 5. Traditional CNN architectures such as VGG16 [18] and InceptionV3 [12] achieved accuracies of 95.8% and 92.3%, respectively, while transfer learning with ResNet50V2 [31] reached 97%. Ensemble-based

Table 5 Result comparison with other studies

Research	Methodology	Accuracy
Abu et al. [18]	VGG16	95.8%
Agrawal et al. [29]	Ensemble Approach	97.32%
Koklu et al. [12]	InceptionV3	92.3%
Hitelman et al. [31]	ResNet50V2	97%
Himel et al. [32]	Ensemble Learning	97.92%
Himel et al. [9]	EfficientNet	97.62%
Our Proposed Model	**ViT**	**98.21%**

approaches further improved performance, with Agrawal et al. [30] reporting 97.32% and Himel et al. [32] achieving 97.92%. EfficientNet, a strong CNN baseline, also produced a competitive accuracy of 97.62% [9].

In comparison, our proposed fine-tuned ViT model surpassed all previous methods, attaining 98.21% accuracy. To the best of our knowledge, this is the first application of ViT on this dataset, demonstrating its ability to capture subtle inter-breed variations more effectively than CNN and ensemble-based approaches. These findings establish ViT as a state-of-the-art solution for sheep breed classification and highlight its potential in advancing smart livestock management.

Ablation study

To better understand the contribution of different design and training choices, we conducted a series of ablation experiments. We systematically tuned epochs, batch size, learning rate, optimizer, and loss function. Results showed that training stabilized around 95–100 epochs, with no significant improvement beyond 100. While accuracy was similar across different batch sizes, a batch size of 32 provided the best GPU utilization. Learning rates above 1e-4 destabilized convergence, whereas a reduced rate of 2e-5 achieved stable and consistent training. Among the tested optimizers (Adam, Adamax, RMSProp), the Adam optimizer combined with categorical cross-entropy produced the best performance.

We evaluated standard augmentation techniques: Random Rotation (± 10–15°), Horizontal Flip (50% probability), Brightness/Contrast Jitter (± 15–20%), Random Zoom/Cropping (90–110% scale), but as shown in Fig. 10, the validation accuracies of all ViT variants remained below the best performance of our non-augmented ViT (98.21%). This indicates that augmentation did not add benefit in our setting, likely because the dataset was already well-balanced and sufficient for training.

To confirm the reliability of our results, we performed 5-fold cross-validation. The accuracies across folds, presented in Table 6, demonstrated consistent performance. ViT-Google achieved the highest mean accuracy (0.9817 ± 0.0007), followed closely by ViT-MAE (0.9767 ± 0.0023), while other models such as ViT-VAN, ResNet50, BEiT, and DiT showed lower but stable results. This consistency across folds reinforces the robustness of our findings and rules out overfitting to a single train-test split.

To further analyze the internal behavior of ViT, we examined the effect of positional encoding and patch size selection. Since ViTs rely on image patches as input tokens, positional encoding is crucial for retaining spatial relationships, while patch size determines the granularity of learned features. We therefore compared performance across different settings, including models trained with and without positional encoding and with patch sizes of 16 and 32. The ablation results in Table 7 show that the default

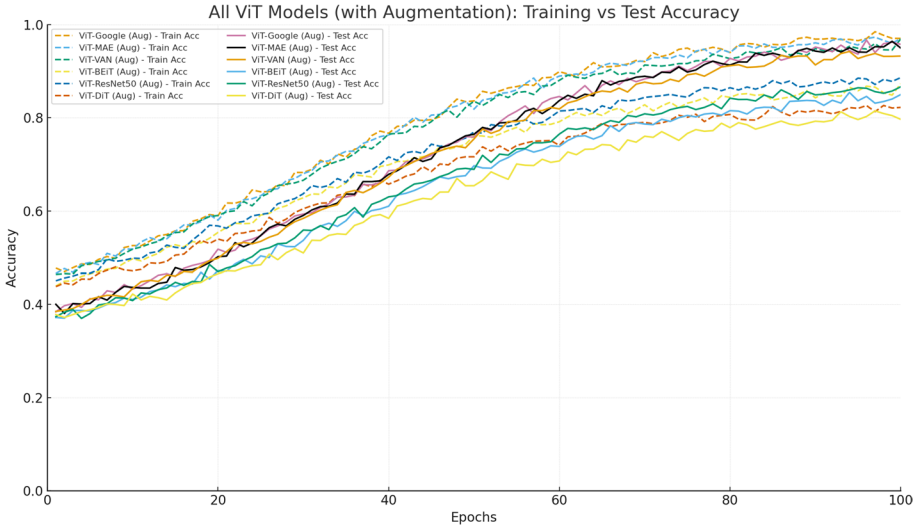


Fig. 10 Effect of data augmentation on ViT models. Validation accuracies remained below 98.21%, showing that augmentation provided no additional benefit beyond the optimized non-augmented configuration

Table 6 5-Fold Cross-Validation results (Accuracy)

Model	Mean Accuracy	Std. Dev.
ViT-Google	0.9817	±0.0007
ViT-MAE	0.9767	±0.0023
ViT-VAN	0.9528	±0.0026
ViT-ResNet50	0.8742	±0.0029
ViT-BEiT	0.8579	±0.0018
ViT-DiT	0.8191	±0.0016

Table 7 Accuracy variance of ViT under different parameter settings

Configuration	Validation Accuracy	Test Accuracy	ROC
Default (Patch = 32, PosEnc ON)	**98.21%**	**98.33%**	0.9987
Patch = 16, PosEnc ON	97.65%	97.82%	0.9974
Patch = 32, PosEnc OFF	96.48%	96.72%	0.9941
Patch = 16, PosEnc OFF	95.83%	95.90%	0.9932

configuration (patch size 32 with positional encoding) achieved the highest accuracy (98.21%). Removing positional encoding or reducing patch size consistently lowered performance, confirming their importance for robust breed classification.

Taken together, the ablation study highlights that while careful hyperparameter tuning is essential, neither data augmentation nor model ensembling was necessary to achieve state-of-the-art accuracy. Instead, the ViT-Google configuration with optimized hyperparameters proved to be the most effective and stable solution for sheep breed classification on a small dataset.

Conclusions

This study demonstrated the effectiveness of ViT for automatic sheep breed classification, achieving state-of-the-art accuracy of 98.21% and outperforming prior CNN and ensemble-based models on the same dataset. By deploying the trained model into the *SheepFormers* mobile application, we bridged research and real-world use, providing

farmers with an accessible tool to support informed breeding and management decisions. The approach is particularly valuable for inexperienced farmers, reducing reliance on costly DNA testing or expert assessment while contributing to the advancement of smart agriculture. Future research should focus on expanding dataset diversity through larger and more heterogeneous image collections, including real-world farm conditions such as varying lighting, occlusion, and wool density. Incorporating federated learning frameworks could enable continuous model updates without centralized data storage, ensuring scalability and adaptability across regions. Additionally, exploring lightweight ViT architectures optimized for mobile devices and conducting usability studies with farmers will further strengthen deployment in practical farming environments.

Appendix A

See Table 8 here.

Table 8 Complete baseline results across all parameter combinations

Model	Optimizer	Loss Function	Learning Rate	Train Loss	Train Accuracy	Valid Loss	Valid Accuracy	Test Accuracy	ROC
Xception	**Adam**	CCE	0.0001	0.16	0.9447	0.4525	0.8335	**0.8333**	**0.96457**
	Adam	KLD	0.0001	0.1748	0.9368	0.5259	0.7996	0.7986	0.95203
	Adamax	CCE	0.0001	0.2465	0.9173	0.5401	0.7849	0.7639	0.94322
	Adamax	KLD	0.0001	0.227	0.9342	0.538	0.758	0.7569	0.94212
	RMSprop	CCE	0.0001	0.2538	0.903	0.4002	0.8512	**0.8472**	**0.96842**
	RMSprop	KLD	0.0001	0.2176	0.9206	0.5453	0.8556	0.8056	0.9573
VGG16	**Adam**	CCE	0.0001	0.4398	0.8737	0.6548	0.7092	0.7292	0.92984
	Adam	KLD	0.0001	0.4464	0.8685	0.6587	0.7451	**0.7431**	**0.93383**
	Adamax	CCE	0.0001	0.6658	0.7852	0.8393	0.6856	0.6806	0.89924
	Adamax	KLD	0.0001	0.6651	0.7975	0.839	0.6906	0.6806	0.89017
	RMSprop	CCE	0.0001	0.434	0.8724	0.7367	0.6788	0.6458	0.93132
	RMSprop	KLD	0.0001	0.4204	0.8698	0.6774	0.7093	0.7083	0.93634
InceptionV3	**Adam**	CCE	0.0001	0.2005	0.931	0.5007	0.7679	0.7639	0.94508
	Adam	KLD	0.0001	0.2352	0.9141	0.5185	0.7799	0.7778	0.9447
	Adamax	CCE	0.0001	0.277	0.903	0.7007	0.7253	0.7222	0.91134
	Adamax	KLD	0.0001	0.3042	0.8958	0.5778	0.7988	**0.7917**	**0.93209**
	RMSprop	CCE	0.0001	0.3426	0.8835	0.6258	0.7565	0.75	0.93737
	RMSprop	KLD	0.0001	0.3758	0.8717	1.2137	0.6799	0.6389	0.98782
Inception-ResNetV2	**Adam**	CCE	0.0001	0.1978	0.9382	0.6899	0.7578	0.7569	0.92438
	Adam	KLD	0.0001	0.1869	0.9375	0.6069	0.8148	**0.8125**	**0.93679**
	Adamax	CCE	0.0001	0.2864	0.9043	0.5807	0.7647	0.7639	0.92348
	Adamax	KLD	0.0001	0.3104	0.8965	0.6579	0.7091	0.7083	0.8993
	RMSprop	CCE	0.0001	0.2831	0.8958	0.6509	0.7951	0.7917	0.94495
	RMSprop	KLD	0.0001	0.3164	0.8854	0.8544	0.745	0.7222	0.89821
DenseNet121	**Adam**	CCE	0.0001	0.0966	0.9707	0.6158	0.7788	0.7778	0.94341
	Adam	KLD	0.0001	0.129	0.9564	0.7891	0.7461	0.7431	0.94772
	Adamax	CCE	0.0001	0.2322	0.9199	0.6419	0.7721	0.7708	0.93788
	Adamax	KLD	0.0001	0.2177	0.9368	0.6039	0.7821	0.75	0.93692
	RMSprop	CCE	0.0001	0.1787	0.9349	0.5078	0.8077	**0.8056**	**0.9528**
	RMSprop	KLD	0.0001	0.1879	0.931	0.9609	0.7449	0.7431	0.93589

Acknowledgements
Not Applicable.
SheepFormers: Vision Transformer-Based Sheep Breed Classification on Small Datasets for Smart Farming.

Author contributions
GMSH contributed to the experimental design, over all methodology and initial manuscript writing. MMI has analyzed the results, re-written the manuscript and organized the experiment & manuscript related task. MMR performed supervision and provided all resources.

Funding
No Funding Available.

Data availability
Data is publicly available in Mendeley Data Repository: https://data.mendeley.com/datasets/cstwjgtxfd/1.

Declarations

Competing interests
The authors declare that they have no conflict of interest. The authors declare that there are no potential conflicts of interest (financial or non-financial).

Received: 14 October 2025 / Accepted: 9 February 2026
Published online: 19 February 2026

References

1. Hossain MS et al (2023) Tissue Artifact Segmentation and Severity Assessment for Automatic Analysis Using WSI, in IEEE Access, vol. 11, pp. 21977–21991. https://doi.org/10.1109/ACCESS.2023.3250556
2. Hossain MS, Shahriar GM, Syeed MMM et al (2023) Region of interest (ROI) selection using a vision transformer for automatic analysis using whole slide images. Sci Rep 13:11314. https://doi.org/10.1038/s41598-023-38109-6
3. Himel GMS, Islam MM, Al-Aff KA (2024) Shams Ibne Karim, Md. Kabir Uddin Sikder, Skin Cancer Segmentation and Classification Using Vision Transformer for Automatic Analysis in Dermatoscopy-Based Noninvasive Digital System, International Journal of Biomedical Imaging, vol. Article ID 3022192, 18 pages, 2024. https://doi.org/10.1155/2024/3022192
4. Sheikh MR (2024) Md. Masudul Islam, Galib Muhammad Shahriar Himel, luffafolio: A multidimensional image dataset of smooth Luffa. Data Brief 110149:2352–3409. https://doi.org/10.1016/j.dib.2024.110149
5. Himel GMS, Islam MM (2024) GalliformeSpectra: A hen breed dataset, Data in Brief, Volume 52, 109996, ISSN 2352–3409, https://doi.org/10.1016/j.dib.2023.109996
6. Market Information Services SHEEP ASSESSMENT MANUAL (2017) Available at: https://www.mla.com.au/globalassets/mla-corporate/prices--markets/documents/minlrs-information-brochures-etc/mla_sheep-assessment-manual_jan-2017.pdf
7. Rowe J, Atkins K (2006) Is the dream of precision sheep production or reality? In: Australian Society of Animal Production 26th Biennial Conference, No. 33
8. Kirton AH, Carter AH, Clarke JN, Duganzich DM (1984) Dressing percentages of lambs. New Z Soc Anim Prod 44:233–233
9. Himel GMS, Islam MM, Rahaman M (2024) Utilizing EfficientNet for Sheep Breed Identification in Low-Resolution Images. Systems and Soft Computing, 1 Apr. pp. 200093–200093. https://doi.org/10.1016/j.sasc.2024.200093. Accessed 4 Apr. 2024
10. Guo Y et al (2023) Sheep Face Image Dataset and DT-YOLOv5s for Sheep Breed Recognition. Computers and Electronics in Agriculture, vol. 211, 1 Aug. pp. 108027–108027. https://doi.org/10.1016/j.compag.2023.108027. Accessed 4 Apr. 2024
11. Ravikiran HK et al (2024) Optimizing Sheep Breed Classification with Bat Algorithm-Tuned CNN Hyperparameters. SN Computer Science, vol. 5, no. 2, 20 Jan. https://doi.org/10.1007/s42979-023-02544-z. Accessed 4 Apr. 2024
12. Koklu M et al Identification of sheep breeds by CNN- based Pre-Trained Inceptionv3 Model. IEEE Xplore, 1 June 2022, ieeexplore.ieee.org/document/9797214. Accessed 31 May 2023.
13. Dosovitskii A et al (2020) An image is worth 16x16 words: Transformers for image recognition at scale. ArXiv Preprint ArXiv:2010.11929
14. Guo et al (2022) Visual attention network. arXiv preprint arXiv:2202.09741
15. He K, Chen X, Xie S, Li Y, Dollár P, Girshick R (2021) Masked Autoencoders are Scalable Vision Learners. arXiv:2111.06377 [cs]. [online] Available at: https://arxiv.org/abs/2111.06377v2
16. Li J, Xu Y, Lv T, Cui L, Zhang C, Wei F (2022) DiT: Self-supervised pre-training for document image transformer. https://doi.org/10.1145/3503161.3547911
17. Bao H, Dong L, Wei F (2021) BEiT: BERT pre-training of image transformers. arXiv:2106.08254 [cs]. [online] Available at: https://arxiv.org/abs/2106.08254
18. Jwade SA, Guzzomi A, Mian A (2019) On-farm automatic sheep breed classification using deep learning. 167:105055Computers Electronics in Agriculture
19. A Noor et al (2020) Automated sheep facial expression classification using deep transfer learning. Computers Electron Agric 175(p 105528). https://doi.org/10.1016/j.compag.2020.105528
20. Spoliansky R, Edan Y, Parmet Y, Halachmi I (2016) Development of automatic body condition scoring using a low-cost 3-dimensional kinect camera. J Dairy Sci 99(9):7714–7725
21. Automated Visual Inspection and Preparation of Live Animals for Meat Processing (2018) Available at: https://www.ampc.com.au/getmedia/703dc062-9000-42b3-ac6b-8337c39aa5f2/AMPC_AutomatedVisualInspectionAndPreparationOfLiveAnimals_FinalReport.pdf?ext=.pdf [Accessed 31 May 2023]
22. Van Hertem T, Viazzi S, Steensels M, Maltz E, Antler A, Alchanatis V, SchlageterTello AA, Lokhorst K, Romanini EC, Bahr C, Berckmans D, Halachmi I (2014) Automatic lameness detection based on consecutive 3D video recordings. Biosyst Eng 119(C):108–116
23. Lu Y, Mahmoud M, Robinson P (2017) Estimation of sheep pain level using facial action unit detection. In: 2017 12th IEEE International Conference on Automatic Face Gesture Recognition (FG 2017), pp. 394–399
24. Sarwar F, Griffin A, Periasamy P, Portas K, Law J (2018) Detecting and counting sheep with a convolutional neural network. pp. 1–6

25. Burke JA et al (2026) An Analysis of the Feasibility of Using Image Processing to Estimate the Live Weight of Sheep. Lincoln University Research Archive (Lincoln University), 1 June 2004. Accessed 2 Jan
26. Carneiro H, Louvandini H, Paiva S, Macedo F, Mernies B, McManus C (2010) Morphological characterization of sheep breeds in Brazil, Uruguay, and Colombia. Small Ruminant Res 94(1):58–65
27. Asamoah Boaheng M, Sam E (2016) Morphological characterization of breeds of sheep: a discriminant analysis approach. Springer Plus 5(1):1–12
28. Salama AE, Hassanien A, Fahmy (2019) Sheep identification using a hybrid deep learning and Bayesian optimization approach' IEEE Access. 7:31681–31687. https://doi.org/10.1109/ACCESS.2019.2902724
29. Agrawal D, Minocha S, Namasudra S, Kumar S (2021) Ensemble algorithm using transfer learning for sheep breed classification, IEEE, in 2021pIEEE 15th International Symposium on Applied Computational Intelligence and Informatics (SACI), Timisoara, Romania, pp. 199–204, May https://doi.org/10.1109/SACI51354.2021.9465609
30. Hitelman A, Edan Y, Godo A, Berenstein R, Lepar J, Halachmi I (2022) Biometric identification of sheep via a machine-vision system. Comput Electron Agric 194:106713. https://doi.org/10.1016/j.compag.2022.106713
31. Himel GMS, Islam MM, Rahaman M (2024) Vision intelligence for smart sheep farming: applying ensemble learning to detect sheep Breeds, artificial intelligence in agriculture. 11 ISSN 2589–7217. https://doi.org/10.1016/j.aiia.2023.11.002
32. Yuan H, Liu Z, Cai Z, Zhang Y, Cheng M (Aug. 2025) Sheep identification based on face features with the method and practical application analysis. Smart Agricultural Technol 12:101362. https://doi.org/10.1016/j.atech.2025.101362
33. Roy M, Medhekar A (Jul. 2025) Transforming smart farming for sustainability through Agri-Tech innovations: insights from the Australian agricultural landscape. Farming Syst p. 100165. https://doi.org/10.1016/j.farsys.2025.100165
34. Phuoc T, Gonario S (Apr. 2025) Climate variability and future responses of agricultural systems in mediterranean region. Farming Syst 100153–100153. https://doi.org/10.1016/j.farsys.2025.100153

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.